



E-Commerce Profit Intelligence Dashboard

Business Insight Summary Report

Project Objective

The objective of this project was to analyze the Global Superstore dataset to understand sales performance, profitability, customer behavior, product performance, and regional market trends. The analysis was performed using Python for data cleaning and preprocessing, MySQL for data validation and exploratory analysis, and Power BI for developing an interactive dashboard that supports business decision-making.

Dataset Summary

- The dataset contains historical sales transactions from the Global Superstore business.
- Key attributes include Orders, Customers, Products, Categories, Sales, Profit, Discounts, Shipping, Markets, and Regions.
- Data cleaning and preprocessing were completed using Python before importing the dataset into Power BI.
- SQL queries were used for data validation and exploratory analysis.

Key Business Insights

1. Sales Performance

- Technology generated the highest overall sales among all product categories.
- Sales showed a consistent upward trend between 2012 and 2015.
- A small number of products contributed significantly to total revenue.

2. Profit Analysis

- Asia Pacific emerged as the most profitable market.
- Technology was the highest profit-generating category.
- Higher discount percentages generally resulted in lower profitability

3. Customer Insights

- The consumer segment represented the largest customer base.

- A limited number of customers generated a significant portion of total sales.
- Customer purchasing behavior varied across different regions.

4. Market Performance

- Regional sales performance differed considerably across markets.
- Asia Pacific consistently outperformed other markets in profitability.
- Certain cities generated substantially higher sales than others.

Dashboard Overview

Page 1 – Executive Overview

Provides a high-level summary of business performance using KPIs, yearly sales trends, category-wise sales, market profitability, and top-selling products.

Page 2 – Sales Analysis

Focuses on monthly sales trends, regional sales, customer segments, shipping modes, top customers, and top-performing cities.

Page 3 – Profit Analysis

Analyzes profit trends, category profitability, market-wise profit, discount impact on profit, and identifies both top profitable and loss-making products.

Page 4 – Customer & Market Insights

Provides customer segmentation, market performance analysis, decomposition tree exploration, top customers, top cities, and executive business insights.

Recommendations

- Increase investment in Technology products due to their strong sales and profitability.
- Review discount strategies to improve overall profit margins.
- Focus customer retention efforts on high-value customers.
- Expand operations in high-performing markets such as Asia Pacific.
- Reassess low-performing products and optimize inventory management.
- Continue monitoring regional performance to identify new growth opportunities.

Tools Used

- Python
- Pandas
- NumPy
- MySQL
- SQL
- Power BI
- DAX
- Git & GitHub

Conclusion

This project demonstrates a complete end-to-end Business Intelligence workflow, beginning with raw data cleaning and validation, followed by exploratory data analysis and the development of an interactive Power BI dashboard. The insights generated help evaluate business performance, identify growth opportunities, and support data-driven strategic decision-making.